CatchUp: Alpha Release and Task Backlog

Zhuoang Tao · CS146S Autumn 2026 · 2026-11-05

Screenshots are from the alpha build (commit `f832bb4`, English digests), cropped for print; larger versions: https://github.com/Zhuoang2/catchup/blob/main/docs/alpha-release.md. "#68" means issue or pull request 68; a date in parentheses after an event points to that day's entry in the AI usage log (`docs/process/ai-usage-log.md`).

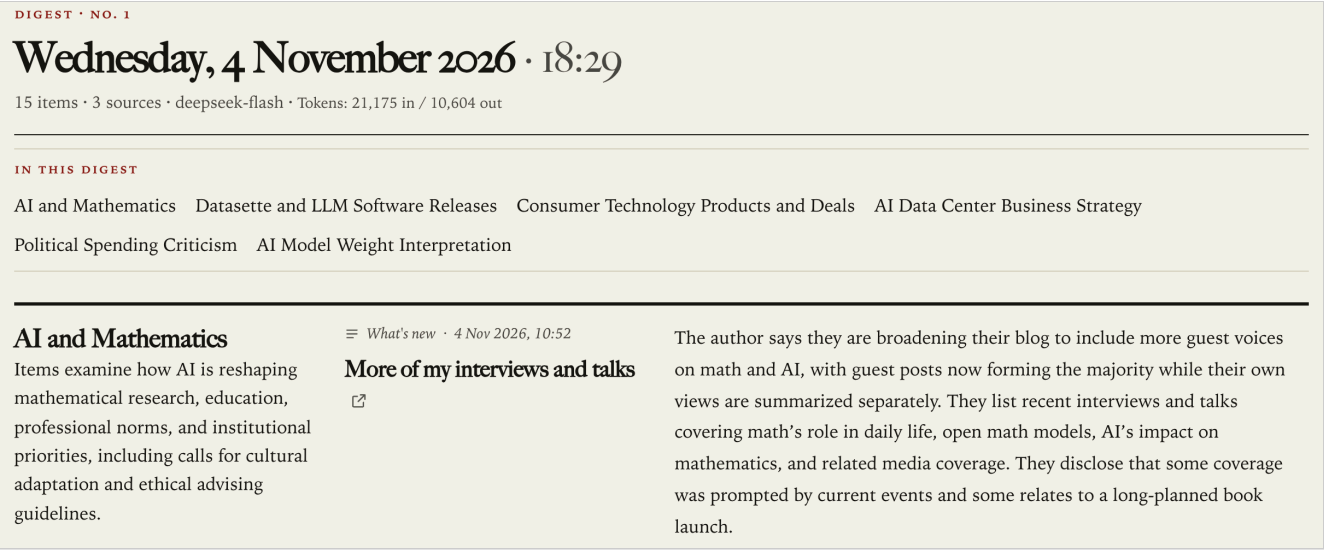
Alpha version

CatchUp is an open-source personal digest that runs on the user's own computer. Users add the sources they follow and connect their own AI model account; **Generate digest** collects everything new since the last check, summarizes and groups it by topic, and links each summary to its original.

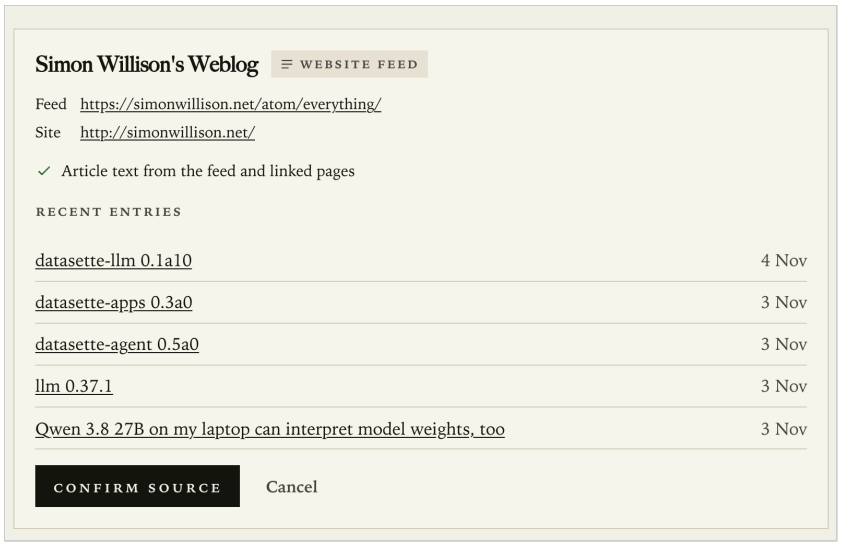
The alpha is commit `f832bb4` (2026-11-03); setup is in the README. To run the alpha, build from the checkout with `docker compose up -d --build`; the published `latest` image is still the October release v0.1.0. 885 backend tests (covering 97% of the backend code), 263 web-app and 114 extension tests need no network or model account; the backend, web-app, and Docker checks must pass before any merge. No screen is a placeholder: unbuilt features are tracked as issues.

Feature	Status	Notes
Model settings	Functional	Services that accept OpenAI's request format (DeepSeek tested); key stored encrypted. Others not started (#12)
Sources: feeds, Bluesky, Mastodon, Reddit, podcasts, X	Functional	Podcasts need a published transcript; X needs the user's own pay-per-use key; Reddit stops its feeds on 2026-11-13 (TechCrunch)
YouTube channels	Best effort	Feeds sometimes fail (one answered 404 on 2026-11-04); YouTube's terms do not allow automated access, so captions are fetched only on opt-in, at the user's risk
Digest run, topics, Latest, History	Functional	Topics often too fine-grained (#32); quality not yet measured (#10)
Linked pages and author replies	Partial	Slow; the progress bar's total can be several times what a run reads (#64)
Images from source sites	Partial	Some chosen images mean little alone (#65)
Browser extension	Partial	Chrome and Edge, installed by hand; may save sponsor text (#62)
Local install and security	Partial	No user accounts; answers only requests addressed to this computer; refuses changes sent from other websites. Known gaps: #68; image addresses that point into the local network (#65). Clean-machine install test pending (#13)
Accessibility	Partial	Contrast and automated checks pass; keyboard focus is lost on some updates (#72)
Bookmarks (#8), top-N (#9), link depth (#54), Bilibili (#40)	Not started	

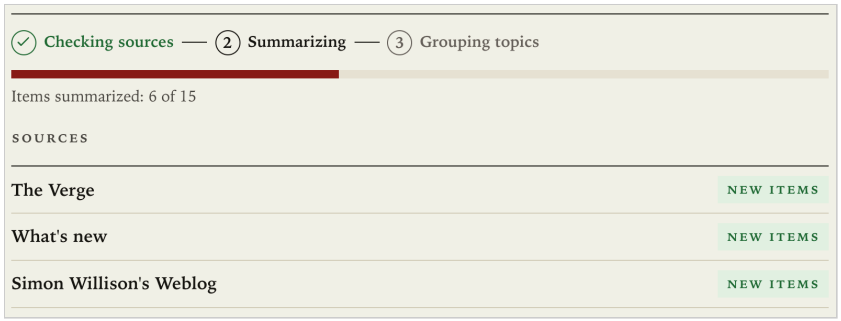
Demo



Latest: the newest digest, its six topics, and the first summary.



Sources: a pasted site is previewed, with its feed and recent entries, before it is saved.





Extension: follow this site, or add this page to the next digest (popup and page captured separately).

A run in progress: the stages, items summarized so far, and a result for each source.

Backlog

The backlog is the public issue list (https://github.com/Zhuoang2/catchup/issues), organized on a private GitHub Projects board, shown here. Each issue's text is its description, usually what happens, why it matters, and the fix; most issues have type and area labels, and the pull request that implements an issue closes it.

	Title	Status	Priority	Labels
1	Summary quality evaluation with real sources #10	Todo	High	area:digest, type:chore
2	Clean-environment install test and setup guide #13	Todo	High	area:deploy, type:chore
3	Test with the user's own sources #14	Todo	High	area:sources, priority:next, type:chore
4	Course: Alpha release, task backlog, AI reflection (due 2026-11-05) #17	In Progress	High	type:course
5	Course: Final release, report, and demo (due 2026-12-10) #18	Todo	High	type:course
6	Pin safe_fetch connections to the address it checked (DNS rebinding) #68	Todo	High	area:sources, priority:next, type:bug
7	Bookmarks, likes, and feedback on digest items #8	Todo	Medium	area:ui, type:feature
8	Group related digest items into fewer, broader topics #32	Todo	Medium	area:digest, type:feature
9	Merge the browser extension's page extractors and stop saving sponsor text #62	Todo	Medium	area:sources, type:bug
10	Make the linked-page pass faster, and show its real progress #64	Todo	Medium	area:digest, area:ui, type:feature
11	Make digest images understandable: captions, and fewer meaningless picks #65	Todo	Medium	area:digest, area:ui, type:feature
12	Items saved during a run's linked-page pass are delivered without their context #70	Todo	Medium	area:digest, type:bug

The project board (table view): the first 12 of the 24 open tasks (all 6 High and 6 of the 10 Medium) with Title, Status, Priority, and Labels, sorted by priority. The older `priority:next` label predates the Priority field.

On 2026-11-04 the board has 57 items: 23 Todo, 1 In Progress (#17, this document), and 33 Done. The 24 open issues are 6 High, 10 Medium, and 8 Low priority. No assignees are used (one-person team).

Not in the screenshot: Medium #71, #72, #73, #76; Low #4, #9, #11, #12, #40, #43, #54, #60. Done tasks have no priority: the field was added on 2026-11-04, after they closed.

The next tasks (their titles are in the board screenshot):

Issue	Description (plain language)	Status	Priority
#68	Stop a followed page from making CatchUp read the user's local network (security)	Todo	High
#13	Install on a clean machine using only the README	Todo	High
#10	Measure summary quality on real sources (target: 90% faithful)	Todo	High
#14	Run CatchUp on the sources the author follows, and record the results	Todo	High
#18	Final release, report, and demo, due 2026-12-10	Todo	High
#70	Pages saved while a run reads linked pages are summarized without their linked pages	Todo	Medium

AI tool integration

Tool	Role
Codex	Product discussion and the proposal (2026-10-01)
Claude Code (Opus 5.5)	Research, plans, UI prototypes and their design reviews, code reviews, live checks, helper agents
Factory Droid (mostly GPT-6 Sol)	Writes the code and tests for most plans, in a separate copy of the repository

How the process works. Context is written once for people and agents: one instruction file for both coding tools (`AGENTS.md`), requirements with 305 example scenarios, and task lists naming each task's test. Most work starts as a written plan: a larger feature gets two proposals for me to choose from (11 of the 26 planned changes so far), and a small feature or fix gets a one-step plan. I approve each plan and each merge, or approve a whole overnight queue in advance. After every Droid run the rules require a bug review (`/code-review`) and a helper agent's requirements review. In about five weeks (2026-10-01 to 2026-11-03) this produced 26 changes and 28 merged pull requests; all 455 non-merge commits on `main` were co-written by Droid or Claude Code. For scale, Droid's first pass over the 14-task interface redesign took about 49 minutes (2026-10-06); review fix rounds followed.

Successes.

- **Security bugs caught.** A review found a malicious web page could make CatchUp send it the stored model key, a flaw in Claude's design (2026-10-03); extension research found any website could start a paid run (#59).
- **Plan review.** A fresh helper agent raised 20 problems in the podcast/YouTube plan, 17 confirmed; an unverified second-round item, tested live, showed some Substack newsletters would go unsummarized (2026-10-05).
- **Overnight work.** On 2026-11-03 I wrote, "I'm going to sleep. Arrange tonight's tasks yourself, for example fix the issues that are open now" (translated). Helper agents implemented five small fixes (PRs #78–#82) while Droid's login was expired; a helper agent's review of #67 caught, before the merge, a plan error that would have let a run look successful even when every model call failed.

Challenges and adjustments.

- **Passing tests did not prove the behavior.** 147 simulated-model tests passed, but the first real DeepSeek run left 4 of 11 summaries empty: its step-by-step "thinking" mode, on by default per DeepSeek's documentation, used up the length limit (2026-10-03). A running-app check is now required, though #66 and #67 had their real-model run only after merging (2026-11-04). Work also became test-first (2026-10-07): "write the task's checks as failing tests and watch them fail, then implement until they pass" (prompt for #59, 2026-10-20).
- **Fix rounds piled up.** I worried that "one patch is patching another patch" (2026-10-21, translated). Of about 75 fix-round items, 18 were real defects, 6 errors in Claude's plans, about 45 tests added only because a deliberately broken copy of the code still passed them (mutation testing), and 2 needless. We then adopted stop rules: sort findings first, one fix round by default, and a fresh helper agent checks each task list.
- **Plan errors.** Droid stopped twice before coding the redesign because the plan contradicted itself, e.g. a color test that the planned colors failed (2026-10-06). The prompt line "stop and report instead of improvising" (since 2026-10-02) kept this cheap, and every numeric check in a plan is now computed before hand-off.

Unexpected insights.

- Most of the 18 real defects were "found by live checks and by reasoning about the code", while mutation testing mostly added test-only checks (decision log, 2026-10-21).
- Plans are cheap to check: a fresh helper agent found 12 false assumptions in the extension plan before coding.